## $^{40}$Ca($\mu^-$,$\nu\alpha$n$\gamma$) **2006Me08**

2006Me08: Muons were generated from the decay of 90-MeV/c $\pi^-$ beam in a 6-m, 1.2-T superconducting solenoid and were selected by a bending magnet at the M9B beamline at TRIUMF. The target was 15-g natural calcium. $\gamma$ rays and x rays were detected using a 186-cm$^3$ HPGe detector. Charged particles caused by electrons from muon decay were tagged by a plastic scintillator in front of the HPGe detector. Measured E$\gamma$, I$\gamma$, E(x ray), I(x ray). Deduced levels, muon capture yields. MeV/c from M9B beamline at TRIUMF. Measured I$\gamma$, $\gamma\gamma$, $\gamma$-p using two HPGe detectors.

Muonic Lyman series for natural Calcium

| $\mu$ x ray | Energy | Intensity in percent |
|---|---|---|
| 2p-1s | 783.659 *25* | 83.8 *10* |
| 3p-1s | 940.63 *10* | 6.2 *2* |
| 4p-1s | 995.48 *10* | 2.0 *1* |
| 5p-1s | 1020.81 *10* | 2.0 *1* |
| 6p-1s | 1034.62 *10* | 1.8 *1* |
| 7p-1s | 1042.71 *20* | 1.4 *1* |
| (8-∞)p-1s | 1046-1063 | 2.8 *4* |

Muonic Balmer series for natural Calcium

| $\mu$ x ray | Energy | Intensity in percent |
|---|---|---|
| 3d-2p | 157.35 *13* | 64.5 *9* |
| 4d-2p | 212.03 *10* | 8.85 *20* |
| 5d-2p | 237.31 *10* | 4.34 *20* |
| 6d-2p | 251.06 *10* | 3.29 *20* |
| 7d-2p | 259.45 *10* | 1.37 *20* |
| (8-∞)d-2p | 261-277 | 1.4 *3* |

### $^{35}$Cl Levels

| E(level)† | $J^{\pi}$† |
|---|---|
| 0 | 3/2$^+$ |
| 1219.3 *5* | 1/2$^+$ |
| 1763.2 *5* | 5/2$^+$ |
| 2645.8 *5* | 7/2$^+$ |
| 2693.7 *5* | 3/2$^+$ |

† From Adopted Levels. Level energies are rounded-off values.

### $\gamma$($^{35}$Cl)

| $E_\gamma$† | $I_\gamma$ | $E_i$(level) | $J^\pi_i$ | $E_f$ | $J^\pi_f$ |
|---|---|---|---|---|---|
| 1219.3 | <0.3 | 1219.3 | 1/2$^+$ | 0 | 3/2$^+$ |
| 1763.1 | 0.5 *3* | 1763.2 | 5/2$^+$ | 0 | 3/2$^+$ |
| 2645.7 | <0.2 | 2645.8 | 7/2$^+$ | 0 | 3/2$^+$ |
| 2693.6 | <0.2 | 2693.7 | 3/2$^+$ | 0 | 3/2$^+$ |

† Rounded-off values from Adopted Gammas.

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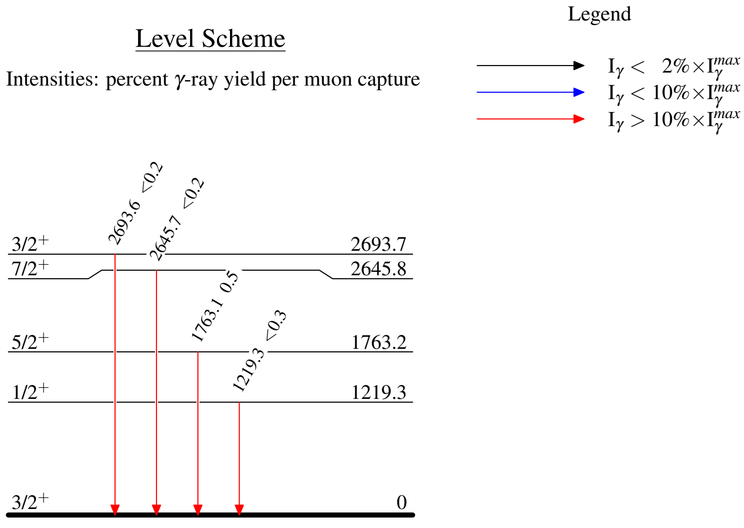
Level Scheme

Intensities: percent $\gamma$-ray yield per muon capture




$^{35}_{17}Cl_{18}$